PEOPLEOPS

HR pipeline & workforce intelligence

A reproducible path from raw HCM exports to attendance controls, payroll signals and CFO actions.

6 MONTHS

1,266 employees

CASE STUDY DELIVERY / APR 01 - SEP 30, 2026

Fictional company; all identities and transactions are synthetic.

One reproducible pipeline, with source-to-output controls

01
RAW LOAD
12 files / row hashes

02
CLEAN
dedupe / map / audit

03
ATTENDANCE
shift-date reconstruction

04
WAREHOUSE
SCD2 / enforced keys

05
ANALYTICS
10 SQL answers

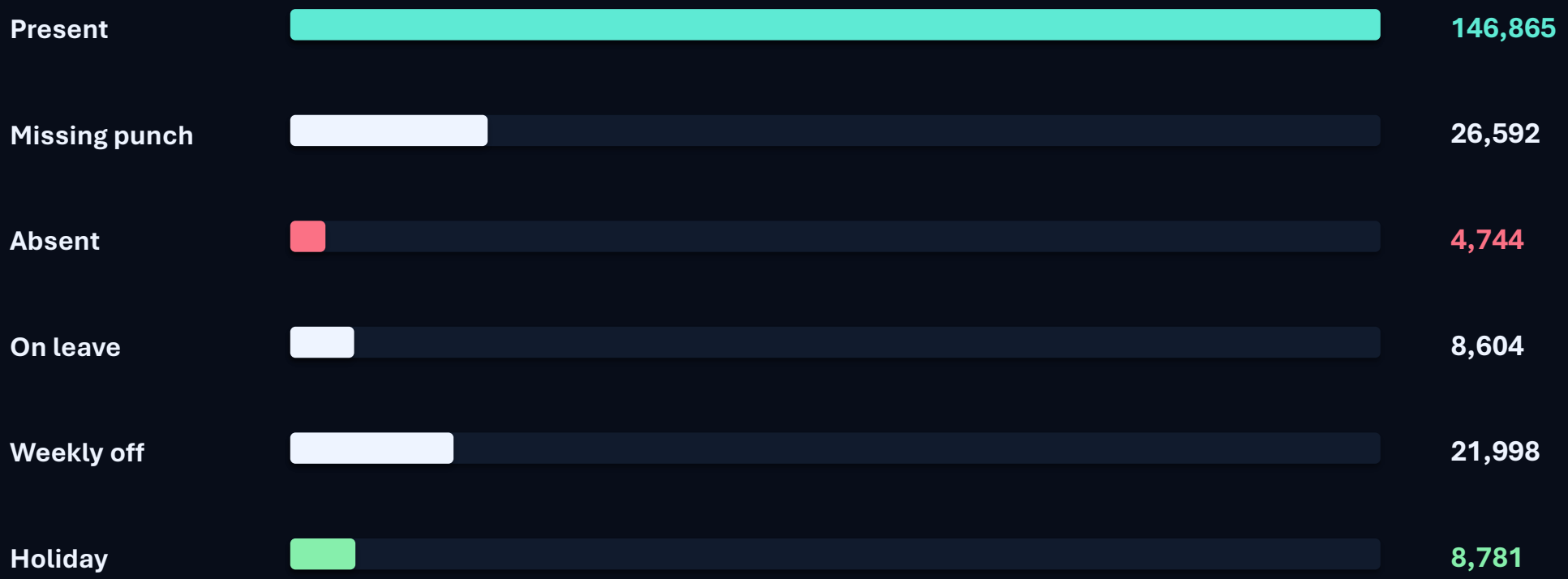
06
QUALITY
reconciliation checks

Automated control coverage

Raw rows accounted for | PK/FK integrity | duplicate employee and punch audit
SCD2 overlap and shift-duration checks | accepted status / leave / payroll values
Final DQ checks failing: 0 | SQLite is permitted by the case-study brief

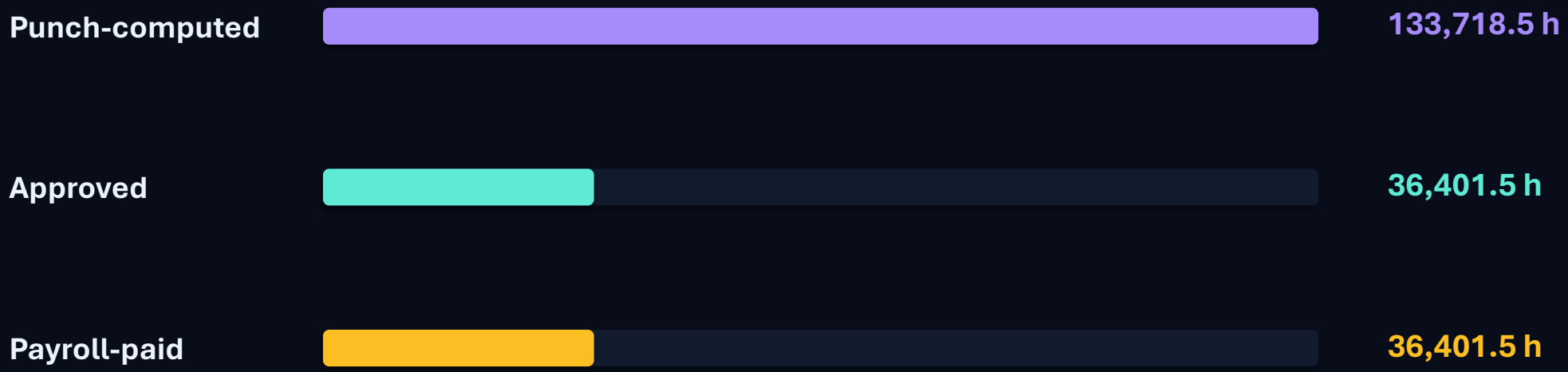
12 source files include 345,280 biometric events and 53,617 payroll lines.

Attendance reconstructed at employee + shift-date grain



217,584 attendance rows | 226,603 late minutes | 2-minute punch dedupe, night-shift start-date, 10-minute grace and scoped holidays

Computed overtime is a reconciliation signal, not an accusation



Gap to paid: 97,317.0 hours | Earnings: PKR 695.0m | Q6 paid OT > computed by 15%: 0 rows

Evidence: q05 by month + department. Reconcile employee/shift dates, approvals and payroll cutoffs.

Three CFO findings, with evidence and an action

97,317.0 h

Punch-computed OT exceeds paid OT

Evidence: q05_overtime_reconciliation.csv

Reconcile at employee + shift date; validate approvals and punch pairs.

26,592

Single-punch employee-days

Evidence: review_missing_punches.csv

Check device uptime, enrollment and repeated patterns before correction.

21

Earnings in months after separation

Evidence: q08_paid_after_separation.csv

Validate final-pay policy, cutoff and last-working-day treatment.

Additional check: 20 annual-leave flags; 59 paid employee-months with zero Present days.

Actions, limitations and the delivered build

01	**OT**	Reconcile computed vs approved vs paid at employee/shift date.
02	**Attendance**	Prioritize single-punch patterns and biometric device reliability.
03	**Payroll**	Validate separation and zero-Present flags against policy.
04	**Data controls**	Review all generated dq_results before CFO sign-off.

Deliverables: reproducible Python + SQLite build | 10 SQL CSVs | dashboard | DQ log | 6-slide deck
Recruitment funnel bonus output included. All discrepancy flags are review signals; the hidden answer key was not provided.